A3 — Feedback Data and Lumped Modeling

Assignment code: A3
Module: PHYS39 Module 6
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1. One-lump model and P-control predictions

The block stores heat while the TEC supplies power and the surroundings remove it:

$$C\frac{dT}{dt} = P_u u - H(T - T_{amb}) \quad \Rightarrow \quad \tau\frac{dT}{dt} = T_{amb} + \chi u - T$$

$$\tau = C/H = 91.69 \text{ s}, \qquad \chi = P_u/H; \quad \chi_h = 0.5311,\ \chi_c = 0.1878\ ^\circ\text{C/PWM}$$

Use χ_h for positive PWM and χ_c for negative PWM. For P control, u = Kp(Tset − T). At steady state:

$$0 = \chi K_p(T_{set} - T_{ss}) - (T_{ss} - T_{amb})$$

$$e_{ss} = \frac{T_{set} - T_{amb}}{1 + \chi K_p}, \qquad \tau_{cl} = \frac{\tau}{1 + \chi K_p}, \qquad L = \chi K_p$$

Subtracting the steady-state balance gives τ d(T − Tss)/dt = −(1 + L)(T − Tss). **L = χKp is the dimensionless loop gain.**

Higher Kp: smaller droop and faster response. P still needs nonzero error to replace heat lost to the room.

Measured comparison: +5 PWM open loop

Part 3 used ambient 24.220 °C and τ = 91.69 s. Prediction: Tss = 24.220 + 0.5311(5).

Predicted temperature	Measured final-minute mean	Prediction error
26.875 °C	26.366 °C	+0.510 °C

The model captures warming but predicts too high a temperature. This trace also supplied τ, so its response time is not an independent check.

Module 5 interpretation

- **Better insulation:** lower H, higher χ, less droop.
- **Double C:** double response time; same steady temperature.
- **Kp = 1:** L = 0.5311 is not much smaller than one. The crossover is Kp = 1.883 PWM/°C.

2. Experimental Goldilocks plots and metrics

All runs used one apparatus, setpoints **28** → **30** → **28 °C**, PWM limit **±255**, anti-windup on, and sample interval ≈0.57 s. Each run lasted ≈93 s: ≈11 s baseline and ≈41 s per step. The integral was reset before each run. Initial temperatures are recorded below; they were not matched.

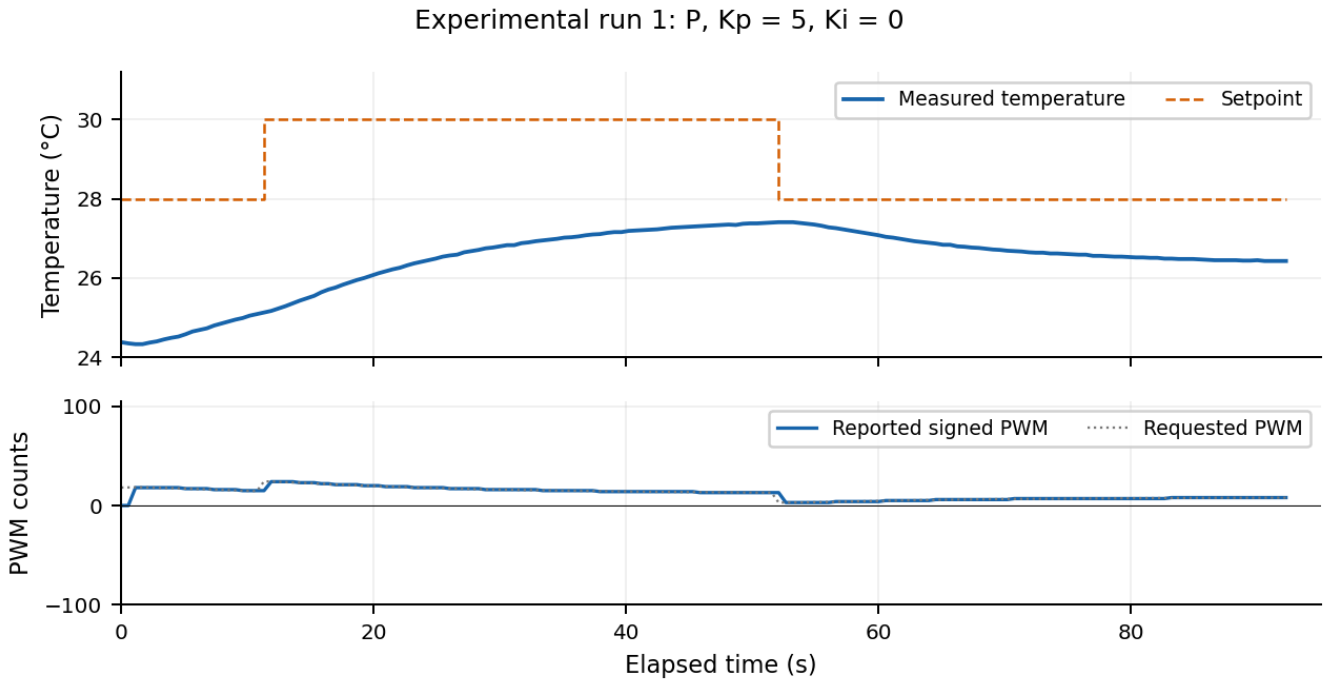


Figure 1. Experimental P, Kp = 5 PWM/°C, Ki = 0. Up/down step starts: 25.09/27.40 °C. Slow response and large error.

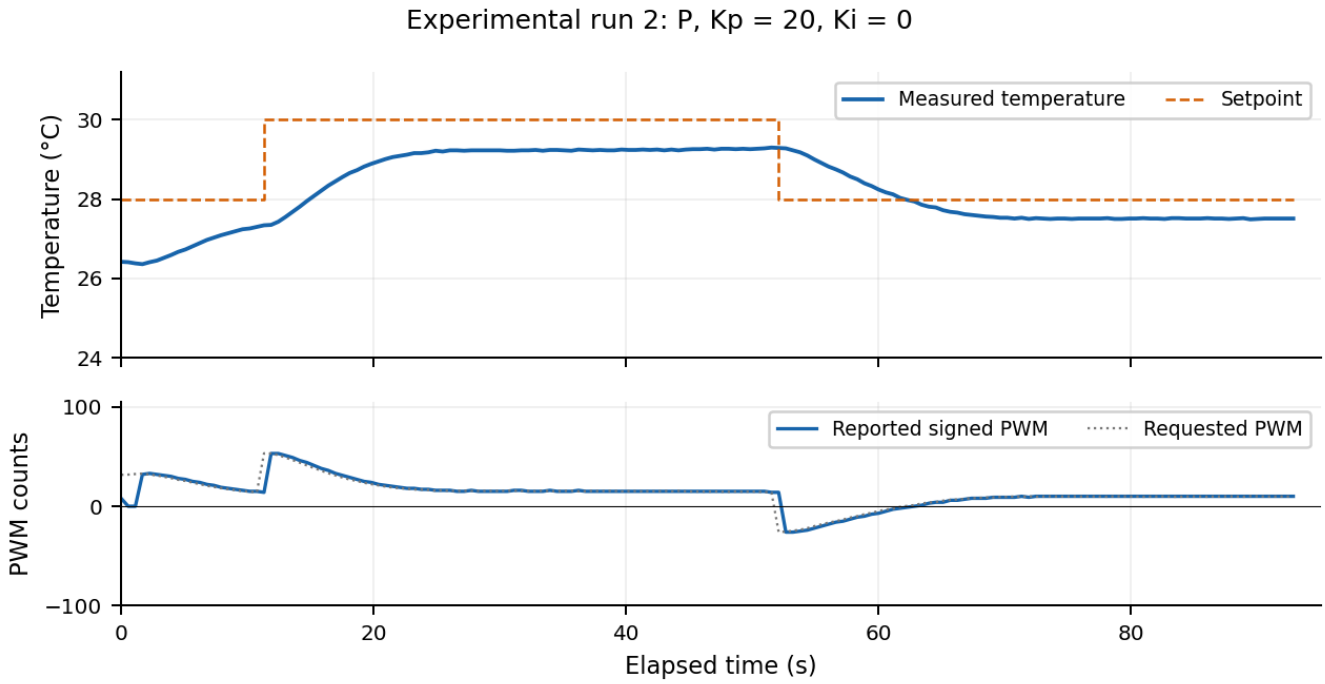


Figure 2. Experimental P, Kp = 20 PWM/°C, Ki = 0. Up/down step starts: 27.30/29.30 °C. Faster response with remaining droop.

Experimental runs 3–4

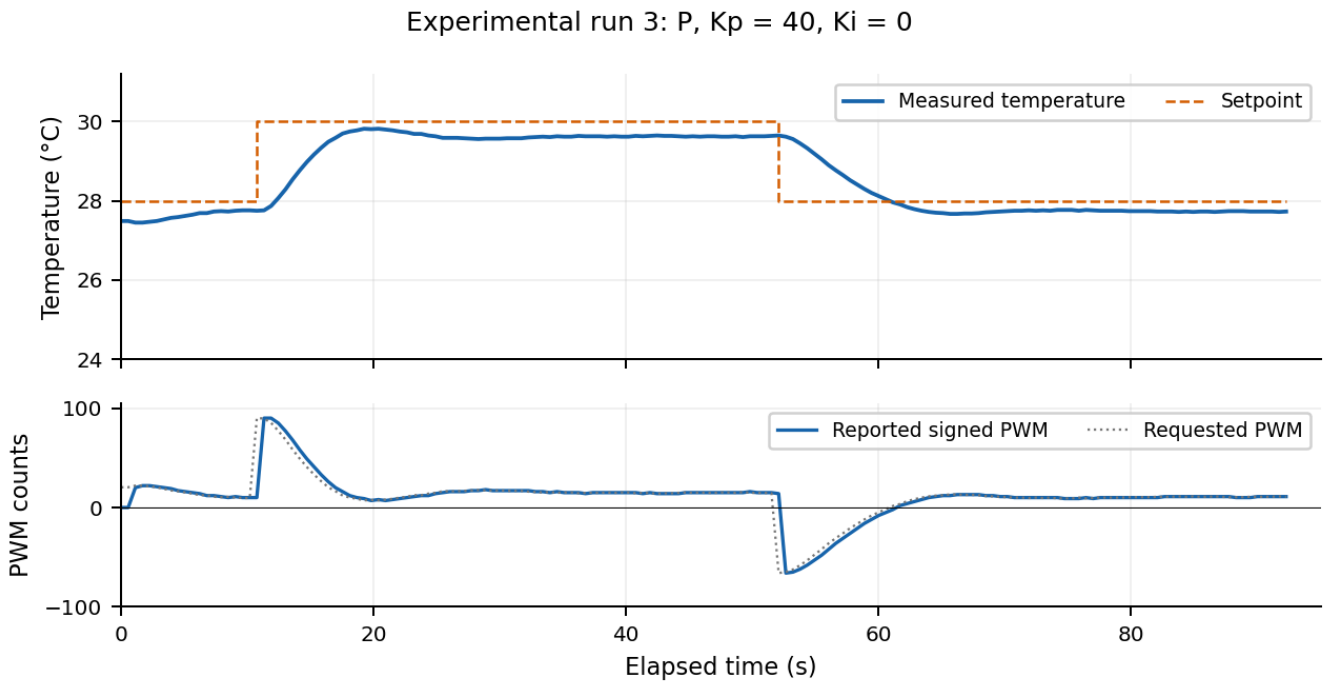


Figure 3. Experimental P, Kp = 40 PWM/°C, Ki = 0. Up/down step starts: 27.76/29.64 °C. Small peak below target; no clear ringing.

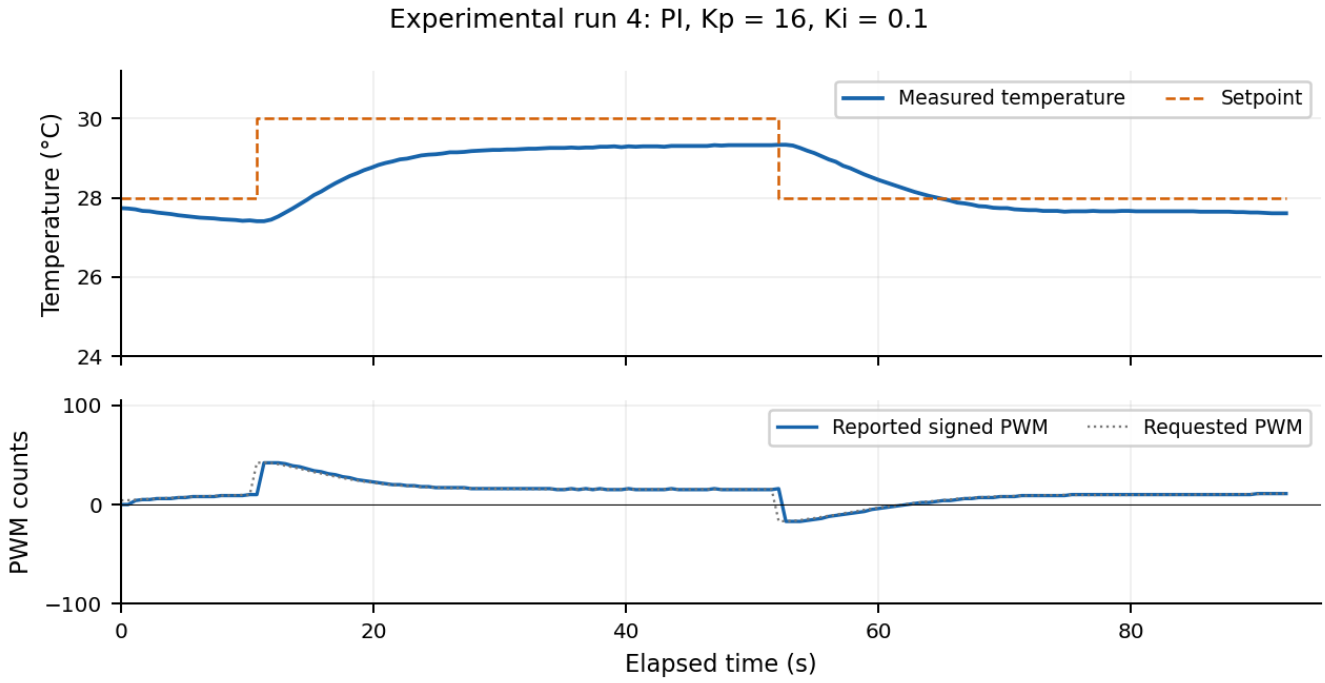


Figure 4. Experimental PI, Kp = 16 PWM/°C, Ki = 0.1 PWM/(°C·s). Up/down starts: 27.43/29.33 °C. Slow integral correction.

Experimental runs 5–6

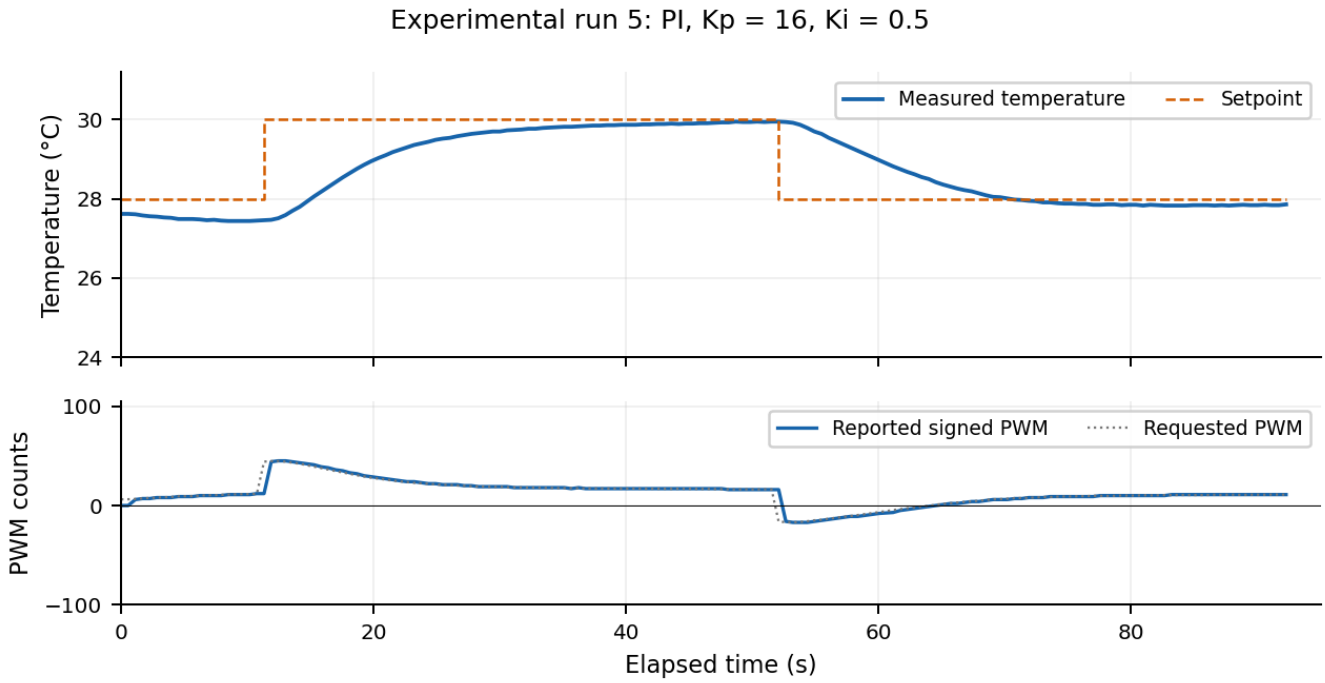


Figure 5. Experimental PI, Kp = 16 PWM/°C, Ki = 0.5 PWM/(°C·s). Up/down starts: 27.45/29.95 °C. Closest tracking with little overshoot.

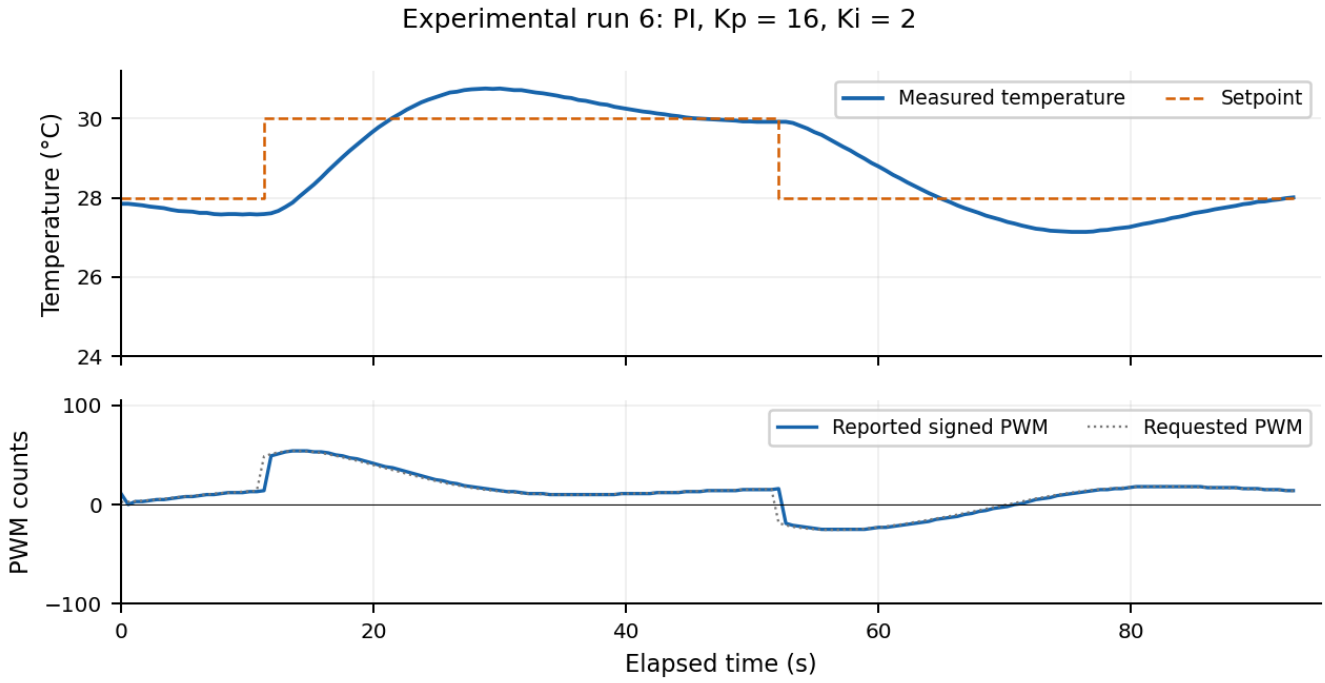


Figure 6. Experimental PI, Kp = 16 PWM/°C, Ki = 2 PWM/(°C·s). Up/down starts: 27.58/29.92 °C. 0.76 °C overshoot and 0.86 °C undershoot.

Largest-gain P and PI: step-response metrics

Rise: first 10%–90% crossing from measured initial temperature toward the new target; linearly interpolated.

Overshoot: excursion beyond the target ÷ 2 °C step × 100%. **Settling:** remain within ±0.04 °C for the rest of the step, with at least 10 s recorded in-band.

Mode Kp/Ki	Step	Rise (s)	Overshoot (%)	End error (°C)
P 40/0	Up	6.40	0.0	+0.373
P 40/0	Down	6.05	16.5	+0.268
PI 16/2	Up	7.01	38.0	+0.058
PI 16/2	Down	9.03	43.0	+0.088

Not established in any of the four windows: settling time, steady-state error, and oscillation period. End error averages the final 5 s; it is not steady-state error.

Saturation: none experimentally; maximum absolute reported PWM was 90. High PI showed one overshoot/undershoot per step.

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3. Two-lump simulations and comparison

One shared plant: χc = 0.1878 and χh = 0.5311 °C/PWM; τ = 91.69 s; ambient 24.220 °C; C = 100.549 J/°C; H = 1.0966 W/°C; Pu,c = 0.20595 W/PWM; G = 10 W/°C; CT/C = 0.35; assumed voltage 12 V; resistance 1.50 Ω.

One shared protocol: both lumps initially 28 °C; targets 28 → 30 → 28 °C at 11 and 52 s; PWM ±255; q(0) = 0; anti-windup on; Δt = 0.025 s. Only mode and gains change.

Model choice: G was adjusted once from 2 to 10 W/°C to reproduce the small high-P peak. G and CT/C are estimates; ambient and τ are reused from Part 3.

Comparison limit: the common 28 °C simulation start differs from the experimental starts, especially at low gain.

$$S = C_T(G + H) + C_m G, \quad \zeta_{2L,P} = \frac{S}{2\sqrt{C_T C_m G(H + P_u K_p)}}$$

$$K_{p,crit} = \frac{S^2/(4C_T C_m G) - H}{P_u}, \quad K_{i,crit} = \frac{S(H + P_u K_p)}{C_T C_m P_u}, \quad \rho_I = K_i / K_{i,crit}$$

At the heating equilibrium, **Kp,crit = 18.46 PWM/°C**; for PI at Kp = 16, **Ki,crit = 8.118 PWM/(°C·s)**. These are local unsaturated criteria. Negative PWM uses the cooling coefficient during downward transients.

Two-lump P simulations

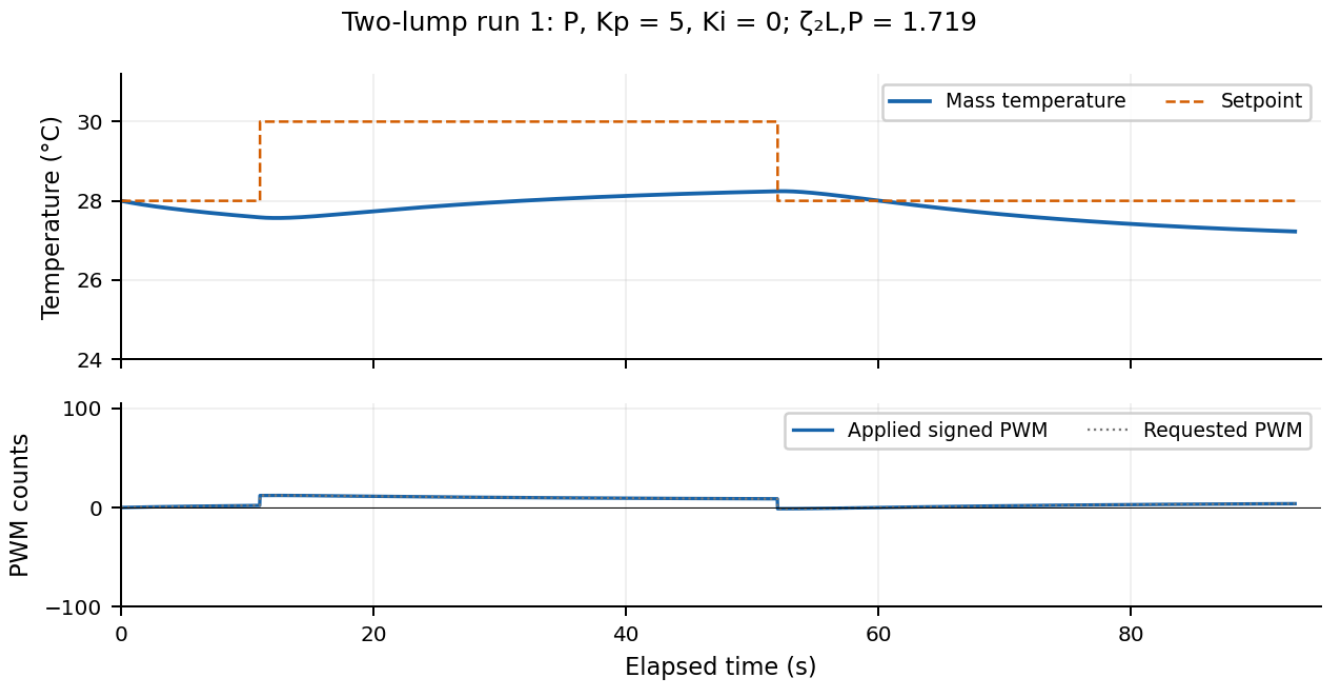


Figure 7. Simulated P, Kp = 5, Ki = 0; $\zeta_{2L,P}$ = 1.719. Stable non-oscillatory.

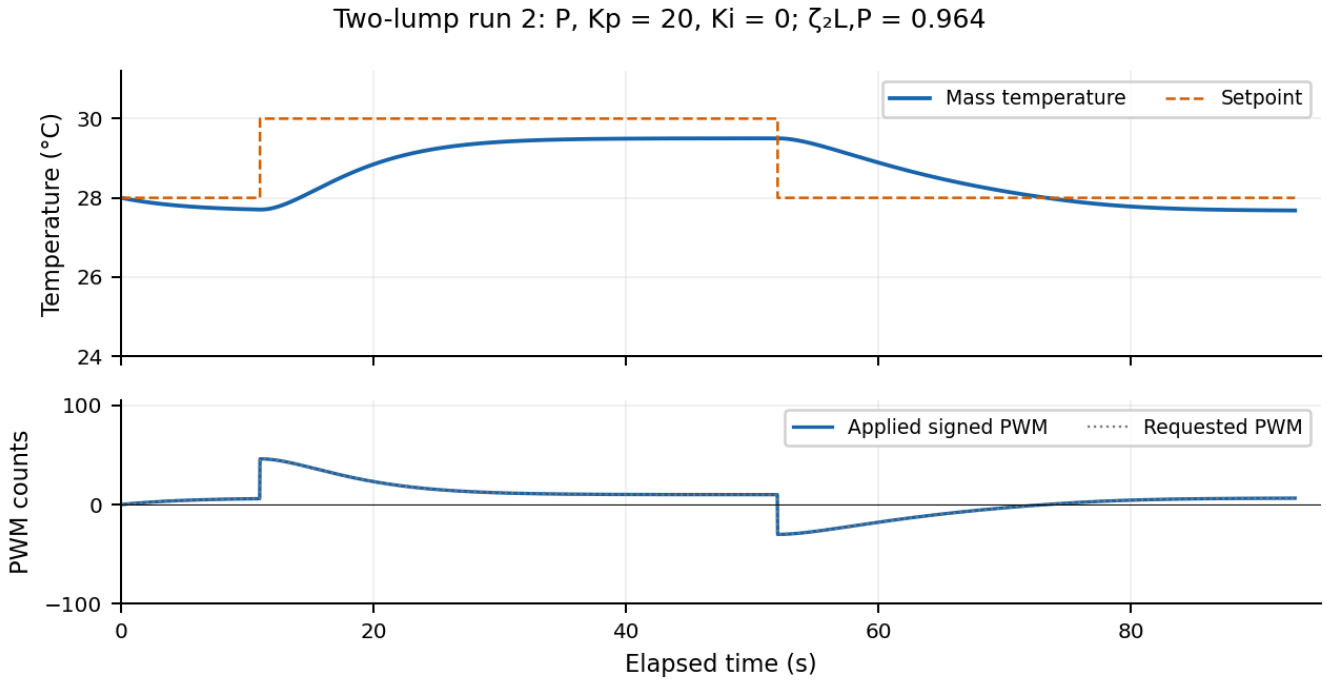


Figure 8. Simulated P, Kp = 20, Ki = 0; $\zeta_{2L,P}$ = 0.964. Stable underdamped.

Two-lump simulations 3–4

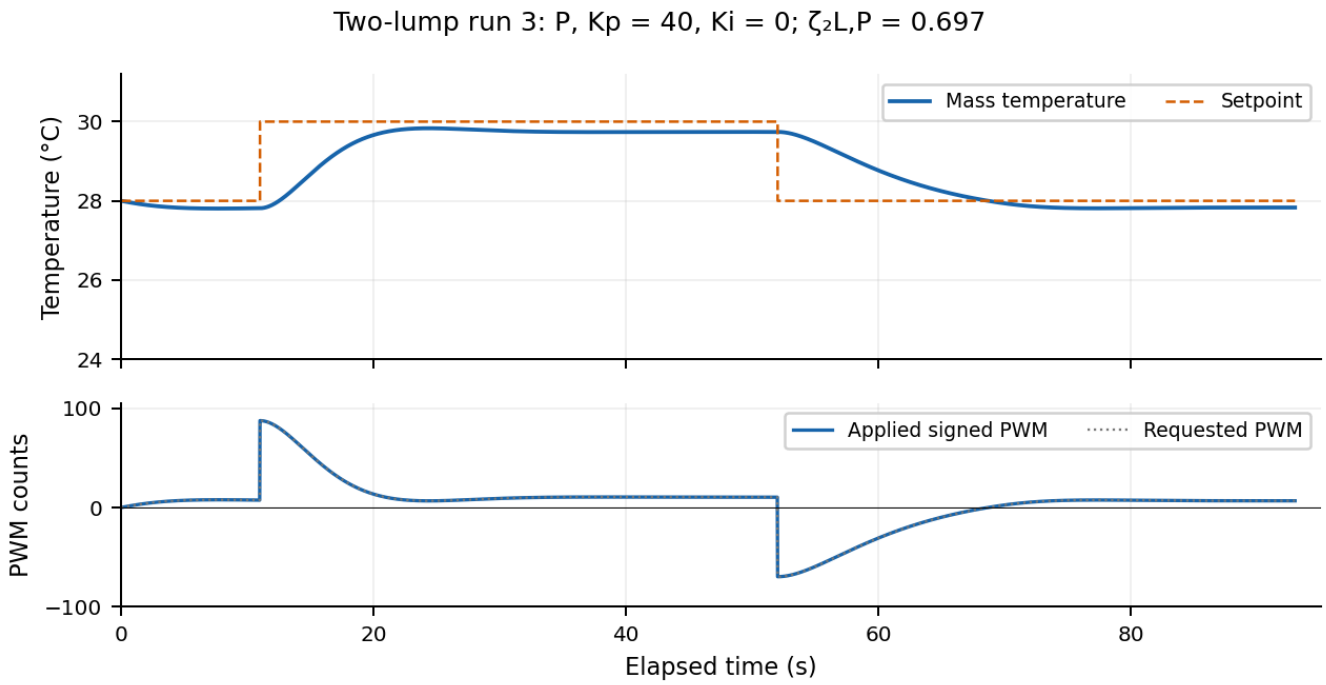


Figure 9. Simulated P, Kp = 40, Ki = 0; $\zeta_{2L,P}$ = 0.697. Stable underdamped.

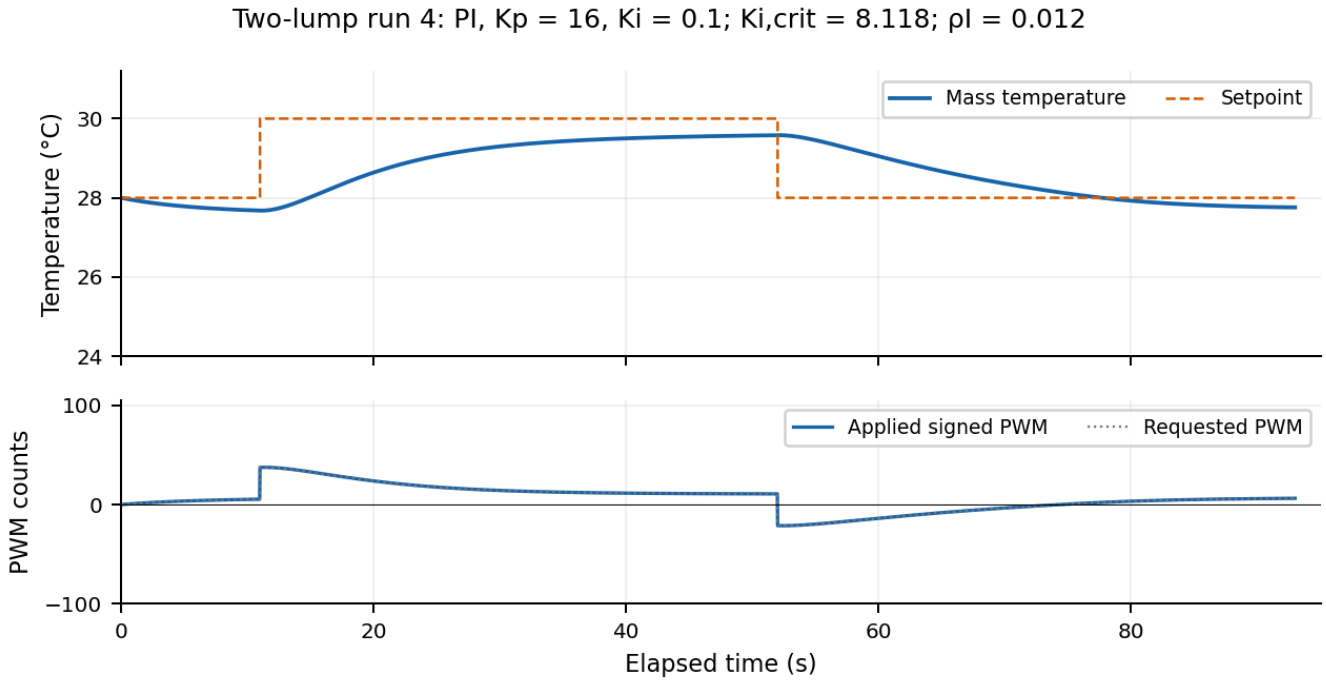


Figure 10. Simulated PI, Kp = 16, Ki = 0.1; Ki,crit = 8.118; ρI = 0.012. Stable non-oscillatory.

Two-lump simulations 5–6

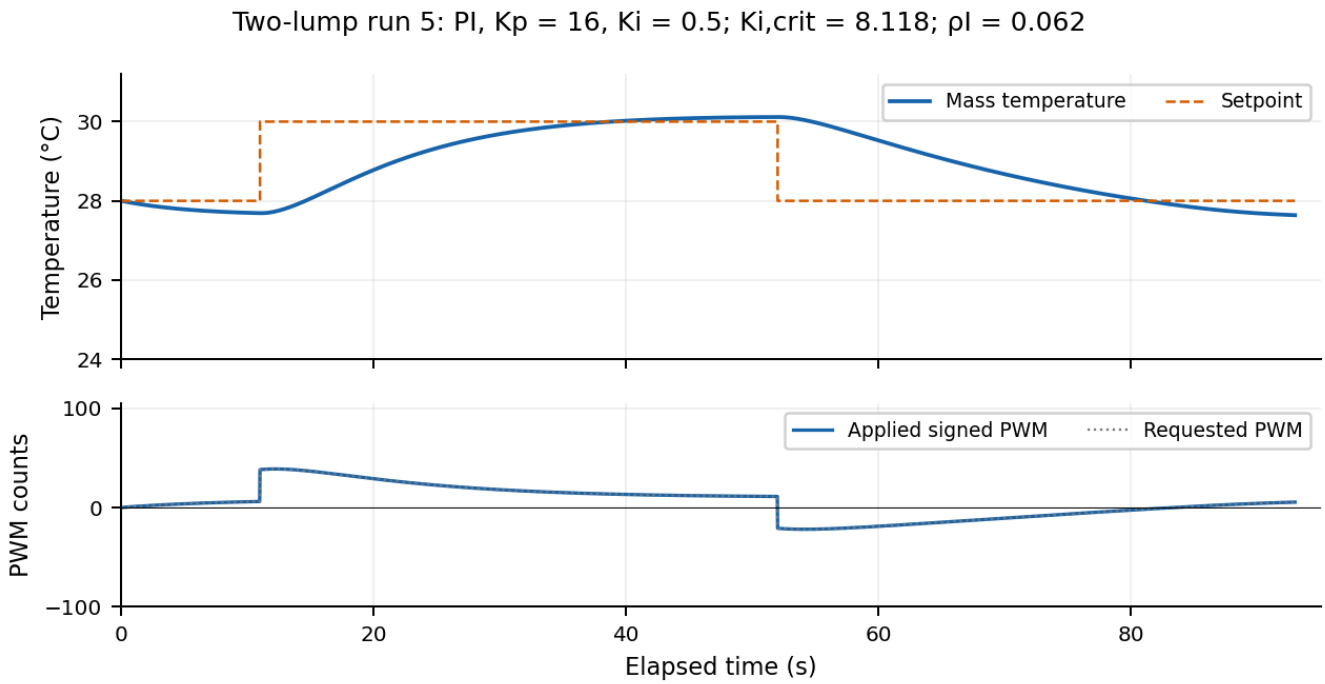


Figure 11. Simulated PI, Kp = 16, Ki = 0.5; Ki,crit = 8.118; ρI = 0.062. Stable non-oscillatory.

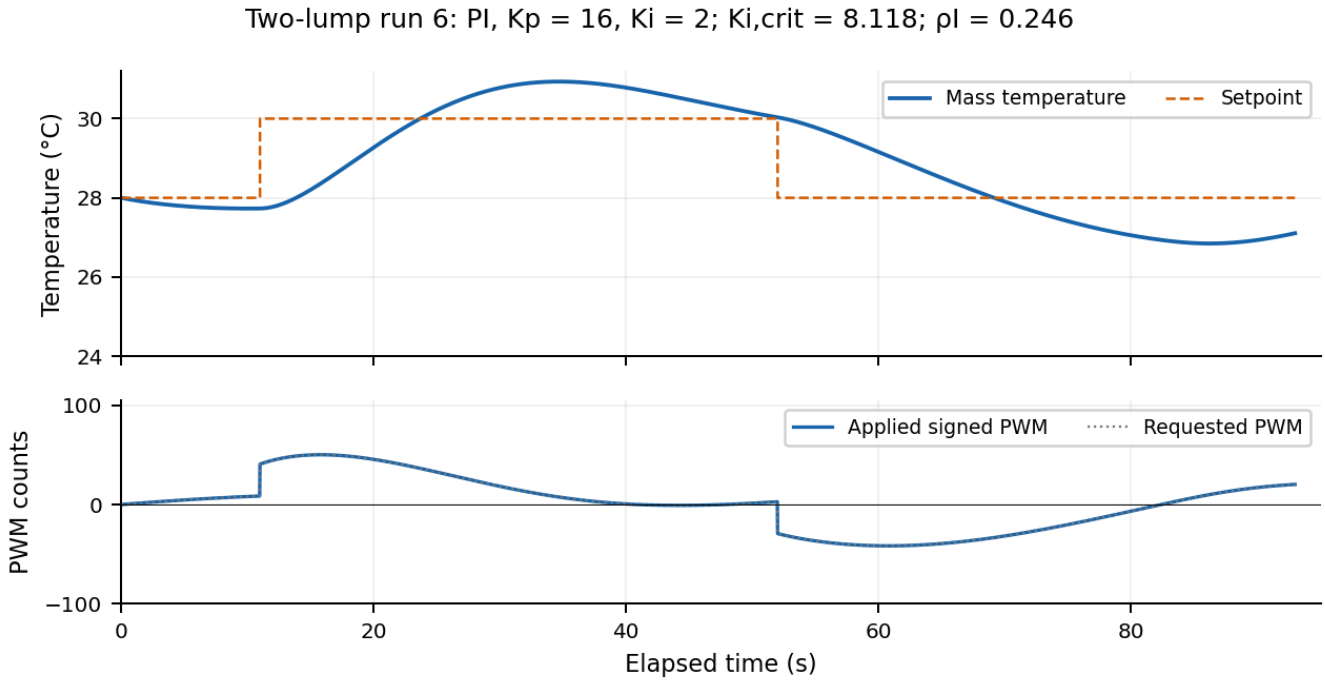


Figure 12. Simulated PI, Kp = 16, Ki = 2; Ki,crit = 8.118; ρI = 0.246. Stable underdamped.

Guided comparison answers

Best matches: high P (run 3) reproduces the small decaying peak below target; intermediate PI (run 5) reproduces the smooth approach close to target. Neither match establishes settling.

P: Kp = 5, 20, 40 lie below, near, and above Kp,crit. Their ζ values are 1.719, 0.964, and 0.697. Higher gain speeds response and lowers damping. All are stable; positive-parameter two-lump P cannot become marginally stable or unstable.

PI: ρI = 0.012, 0.062, 0.246: all stable. Runs 4–5 have real poles; run 6 is underdamped. Real-pole PI can still overshoot because q starts away from equilibrium.

Increasing Ki strengthens correction but increases ringing. Marginal stability is at ρI = 1; instability is above 1. Our retained gains stay below this boundary.

Agreement: stronger PI correction reduces early error, then produces overshoot.

Comparison (up/down)	Experiment	Simulation
High-P rise time	6.40 / 6.05 s	9.05 / 11.53 s
High-PI overshoot	38 / 43%	46.8 / 57.7%

Disagreement: exact speed and overshoot do not match. Different starting states and estimated internal parameters limit agreement. No recorded growing or sustained oscillation contradicts the predicted stability in these windows.

Controlled checks

• **One-lump PI:** higher Ki lowers ζ; higher Kp raises ζ; doubling C divides ζ by √2.
• **Two-lump practice:** Kp = 60, 105, 160 gives ζ ≈1.31, 1.00, 0.81. Raising G from 35 to 70 increases damping; a more equal mass split lowers damping.

Simulated saturation: the Ki = 3 practice run comes within 0.01 PWM of ±255 at 191.8 s. Early growth shows linear instability; later output limiting and anti-windup bound the response. This was not experimental saturation.

Data, code, and modeling checkpoint

Experimental sources: six October 14 CSVs in the Part 7 data directory. Figures plot reported signed PWM alongside requested PWM; telemetry reports the previous interval's output. Simulation data, metrics, and calculations are reproducible with build_a3.py. Halving simulation Δt changes temperature by less than 0.0022 °C in the six retained runs.

Local files: analysis/build script; experimental metrics; settings and stability calculations.

Raw recordings: Run 1 raw CSV · Run 2 raw CSV · Run 3 raw CSV · Run 4 raw CSV · Run 5 raw CSV · Run 6 raw CSV

GitHub modeling checkpoint: 73d29c9 (existing Part 3/4 checkpoint). This checkpoint predates the six runs and A3; the new A3 files have not been committed or pushed.

Assignment: Lab 6 — A3. Two-lump protocol: Lab 7 — six simulation runs.

4. PI steady state, assumptions, and windup

$$\dot{q} = e = 0 \Rightarrow T_{ss} = T_{set} = 55\ ^\circ\mathrm{C}, \quad u_P = 0, \quad u_I = H(55 - T_{amb})/P_u$$

Ideal result: 55 °C. At equilibrium q is constant, so e = 0. The integral retains the heat-loss-balancing output while P becomes zero.

Required heating command: (55 – 24.220)/0.5311 = **57.95 PWM**, within ±255 in this model. This is a prediction, not a 55 °C experiment.

Conditions: stable feedback, sufficient heating power, no persistent saturation or shutdown, Ki > 0 with continued integration, and accurate/timely measurement. Sensor bias shifts the actual temperature away from the indicated 55 °C.

Windup: stored error keeps growing while PWM is capped, causing overshoot and slow recovery.

Anti-windup: reject updates that push farther beyond the limit; accept updates that bring the request back.

Reset: clearing q removes the balancing heat command. Temperature initially falls while I rebuilds.

5. Model limits and gain selection

Complete this question only after completing Part II of Module 6.

$$T(t) - T_{ss} = [T(0) - T_{ss}]\exp[-(1 + \chi K_p)t/\tau]$$

One-lump limit: the positive, decaying exponential cannot cross its own steady temperature. From below the steady temperature and target, P cannot overshoot the target or ring.

$$C_T\dot{T}_T = P_u u - G(T_T - T_m), \qquad C_m\dot{T}_m = G(T_T - T_m) - H(T_m - T_{amb})$$

Two-lump lag: the TEC heats its own mass first; G then transfers heat to the measured mass. Stored TEC energy keeps changing Tm after the command falls, allowing overshoot and decaying oscillations.

Selected P gain: Kp = 20 PWM/°C is close to Kp,crit = 18.46, with ζ = 0.964: fast response near critical damping. It retains droop; gain 40 reduces droop further but has a more pronounced transient peak.

Selected PI gains: Kp = 16 PWM/°C and Ki = 0.5 PWM/(°C·s), with ρI = 0.062, well below 1. Run 5 approached both targets with little overshoot. Ki = 0.1 corrected too slowly; Ki = 2 produced larger excursions. These are provisional choices because the recordings did not establish full settling.

Next model improvement: include the ≈0.57 s measurement/command delay. Temperature-dependent TEC coefficients and thermal contact changes may also matter.